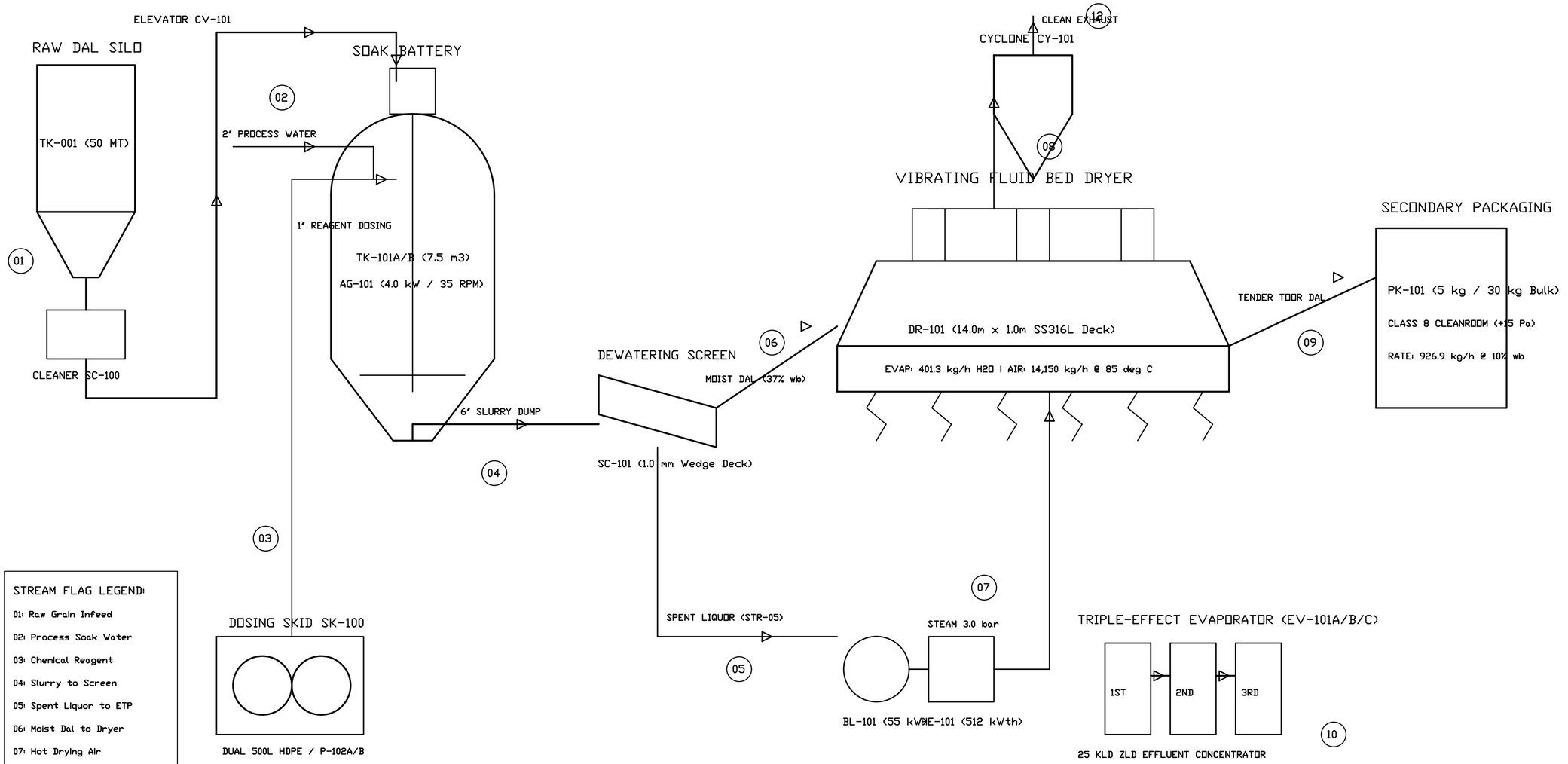
KEY PROCESS DESIGN BASIS & THERMODYNAMIC OPERATING PARAMETERS:

Throughput: 15,000 kg/day as-fed Dal | Operating Schedule: 16.0 h/day (Mean 937.5 kg/h, Design 1,102.9 kg/h @ 85% OEE)

Infeed Moisture: 10.50% w.b. (839.06 kg/h dry solids) | Product Moisture: 10.00% w.b. (926.90 kg/h saleable dal) | Solids Yield: 99.40%

Soak Liquor Ratio: 1.50 : 1.00 (Water:Dal) | Dosing Reagent: Food-Grade NaHCO3 / Citrate | Drying Air: 85 deg C | Deck Area: 14.0 m2

All Product-Contact Metallurgy: AISI SS316L 2B Sanitary Polish (Ra <= 0.8 um) | CIP / Washdown Compatible | ZLD Effluent Evaporation




STREAM FLAG LEGEND:

01: Raw Grain Infeed

02: Process Soak Water

03: Chemical Reagent

04: Slurry to Screen

05: Spent Liquor to ETP

06: Moist Dal to Dryer

07: Hot Drying Air

09: Saleable Toor Dal

| | | |
|---|---|---|
| FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.<br>15,000 kg/day Split Pigeon Pea Optimization Greenfield Project | | |
| PROCESS FLOW DIAGRAM - OVERALL PLANT<br>DISCIPLINE: PROCESS | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP) | |
| DWG NO: DWG-FIN-15T-001 | REV: V8.0 | SHEET: 1 OF 12 |
| SCALE: AS SHOWN DATE: 2026-10-17 | STATUS: PRELIMINARY | GATE: G-A HOLD |

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ